

DDA5001 Machine Learning

Lec 1: Introduction

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About the Course

Machine Learning

Machine Learning

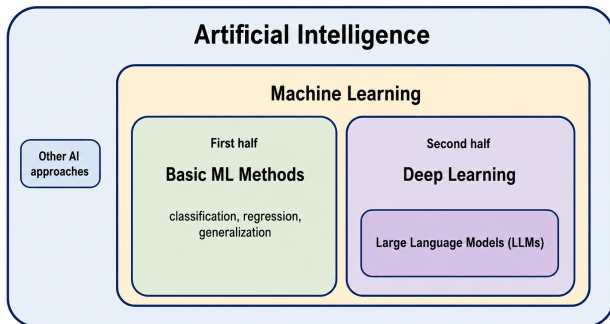
Learn from data to train models. These models can uncover structure, make predictions, generate content, or support decisions.

Machine learning is particularly useful when:

- ▶ The rules are difficult to specify explicitly, but examples or data are available.
- ▶ Predictions or decisions must be made at scale or adapted to changing data.
- ▶ We want to discover hidden structure or generate new content from data.

AI, Machine Learning, and LLMs

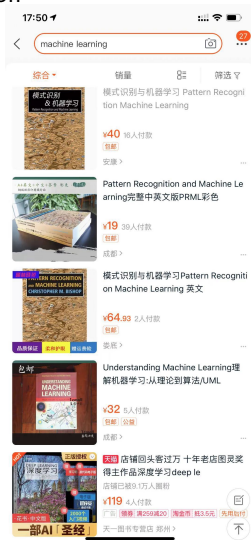
- ▶ **Artificial intelligence (AI)** is the broad field of building systems that perceive, reason, generate, plan, or act. **Machine learning is the dominant approach in modern AI.**
- ▶ This course first covers **basic ML methods**, then moves to **deep learning** and **large language models (LLMs)**.



Motivations: Machine Learning is Ubiquitous

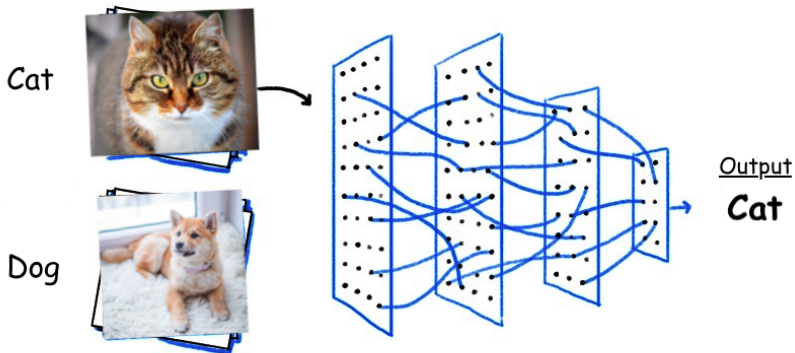
Ubiquity of Machine Learning

► Product recommendation



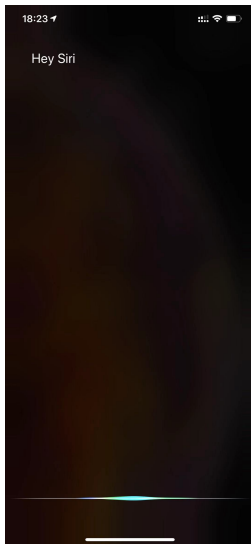
Ubiquity of Machine Learning

► Image classification



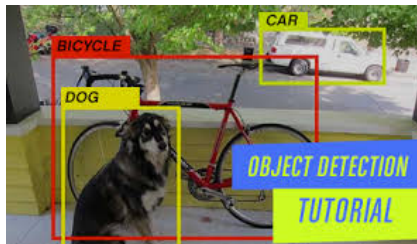
Ubiquity of Machine Learning

► Speech recognition



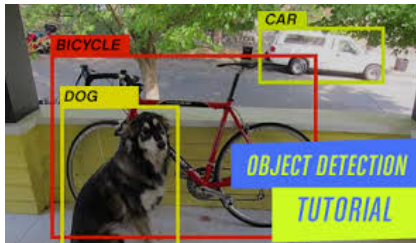
Ubiquity of Machine Learning

► Object detection



Ubiquity of Machine Learning

- Object detection

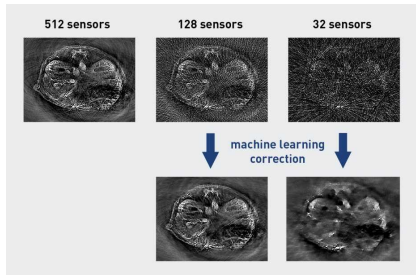


- Autonomous driving



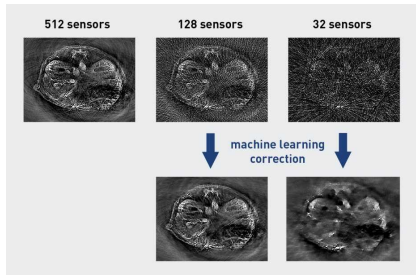
Ubiquity of Machine Learning

► Medical imaging



Ubiquity of Machine Learning

► Medical imaging

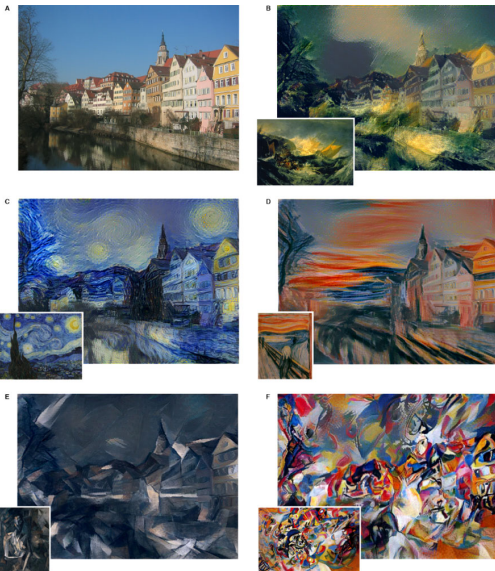


► Health care



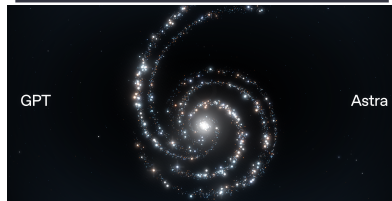
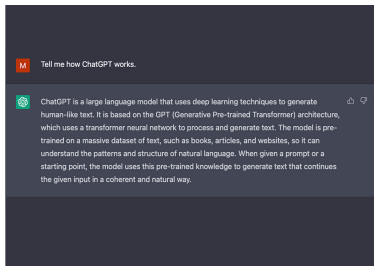
Ubiquity of Machine Learning

- 'Artificial artists'— AIGC (generative AI)



Ubiquity of Machine Learning

► GPT (Large language models)



▪

Course Information

Lecture and Instructors

Lectures

► Wednesday 10:30 - 11:50 a.m., Zhixin 111.

Instructor: Prof. Junchi Yang

Email: yangjunchi@cuhk.edu.cn

Office: Daoyuan 321A

Office hour: Wednesday 2:00 - 3:00 p.m.

Home page: junchi-michael-yang.github.io

Research interests: Optimization theory and machine learning

Course Content

What this course will be?

- ▶ **Methodologies:** The fundamentals and methodologies of machine learning, including MLE, least squares, training versus testing, logistic regression, overfitting, gradient-based algorithms, PCA, neural networks, deep learning, transformers, etc.
- ▶ **Applications:** Design and apply machine learning methods (including model and algorithms) to solve practical problems, e.g., large language models.

After studying this course, the students are expected to be able to apply machine learning algorithms in **industry** and be well prepared in conducting advanced machine learning **research**.

Course Outline and Plan

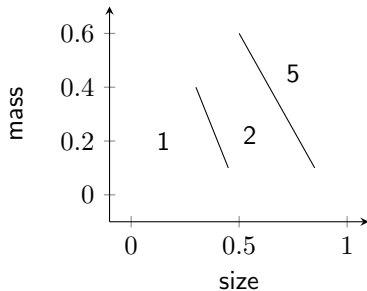
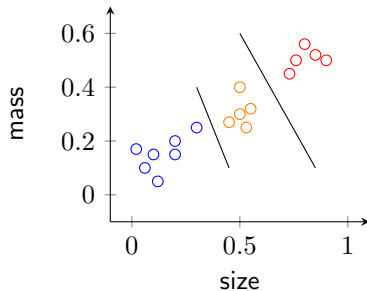
The main line of this course covers two main types of learning:

- ▶ **Supervised Learning** (main)
 - ▶ Linear models
 - ▶ Training versus testing
 - ▶ Overfitting
 - ▶ Training algorithms
 - ▶ Nonlinear models
 - ▶ Advanced topics such as optimizers, transformers, and language models
- ▶ **Unsupervised learning** (a very small part)
 - ▶ Dimensionality reduction
 - ▶ Clustering

Supervised Learning

Example from vending machine — coin recognition:

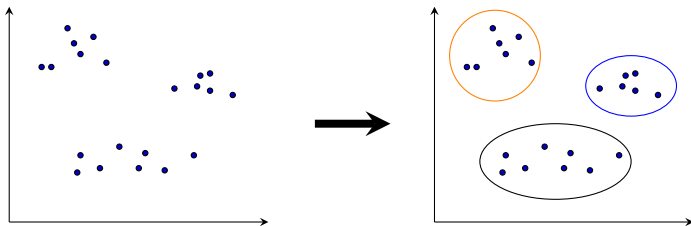
input data + label



Unsupervised Learning

Instead of input + label, we only have:

input data



Prerequisites

Machine learning lies in the intersection of several disciplines and its needs fundamental mathematical knowledge.

To study machine learning, we need:

- ▶ Probability and statistics;
- ▶ Multi-variate calculus (mainly calculation of the gradient);
- ▶ Linear algebra (vector, matrix, norm, subspaces);
- ▶ Optimization (algorithms).

We will go through basic mathematics later in the next lecture. These math notions are important. For instance, linear algebra is the basic “language” of this course, we have to represent all the data and operations in vectors and matrices.

Tutors

We have 8 tutors in this semester:

- ▶ KONG Mingze
225040539@link.cuhk.edu.cn
- ▶ SUN Jiahang
225045021@link.cuhk.edu.cn
- ▶ SHANG Zhiwei
225040503@link.cuhk.edu.cn
- ▶ LU Pingchen
2720801657@qq.com
- ▶ QU Zikun
225040504@link.cuhk.edu.cn
- ▶ SHEN Jiarun
225045001@link.cuhk.edu.cn
- ▶ LI Mengqi
223040115@link.cuhk.edu.cn
- ▶ LIU Yanwen
225040486@link.cuhk.edu.cn

Their office hours are by appointment.

Tutorials

We will have **4 tutorials**. Note that these tutorials **do not** take place every week. The tentative plan for tutorials is:

- ▶ Tutorial 1 is in the 2nd teaching week (Sep 15). Main contents: install and configure **OpenCode and Codex**, and introduce their key capabilities, including code generation, debugging, repository navigation, and agentic task execution.
- ▶ Tutorials 2-4 are about our **course project** and its **large language model** components. They will take place in the 9th teaching week (Nov 10), 11th teaching week (Nov 24), and 13th teaching week (Dec 8), respectively.

There will be no separate tutorials on Python, individual learning algorithms, or homework walkthroughs. **Students are expected to use coding agents (but not simply copy-paste)** to resolve routine technical and assignment-related questions independently, subject to the course's academic honesty policy.

The tutorial venue and time are:

- ▶ **6:30pm - 7:30pm, online**, the zoom link will be provided.

Grades

The evaluation of this course contains three parts:

- ▶ Homework (18%)
- ▶ Course project (30%)
- ▶ Final exam (52%)

Homework (18%)

We will have about 3 homework.

- ▶ You will have two weeks for completing each assignment.
- ▶ Through completing the homework, you are supposed to learn how to apply machine learning methods to practical problems. It includes programming (Python) and algorithm implementation (more than call software package).
- ▶ There will be no separate tutorials on Python, individual learning algorithms, or homework walkthroughs. Use coding agents for routine technical and assignment-related questions, subject to the academic honesty policy.
- ▶ Learn how to work together with AI, which will be your future working style. NOT just copy-paste whatever AI outputs. You should understand AI's outputs, and learn how to master AI to give you even better outputs.
- ▶ Late submission will not be graded.

Course Project (30%) I

We will have a course project.

Format:

- ▶ It is about Large Language Models (LLMs).
- ▶ The project is cut into 3 consecutive parts. Parts I and II each have 2 weeks to complete. Part III has 3 weeks, including the final review and exam period. Thus, the project will last 7 weeks in total.
- ▶ The project will be released around the 8th week (Nov 8).
- ▶ Individual & Group work (2 students to form a group; you can also work individually). The first part is individual work, while the remaining two parts are group work. For group work, each member in a group should clearly state which part you are taking in charge. Find a suitable group by yourself in this semester (the best way we figured out in the past).

Course Project (30%) II

Submission materials: Assessment will be conducted according to the following materials:

- ▶ A PDF report for part I (up to one-page), a PDF report for part II (up to two-page), and a PDF report for part III (up to two-page). These page limits do not contain references and appendix pages.
- ▶ Code.

Project grade:

- ▶ The project contains 100 points. The distribution is: part I: 30, part II: 40, part III: 30.

Project tutorial: We will have 3 project tutorials (Tutorials 2-4) to explain large language models and provide guidelines on how to complete the project. They will take place in the 9th teaching week (Nov 10), 11th teaching week (Nov 24), and 13th teaching week (Dec 8), respectively.

Final Exam (52%)

- ▶ Final exam covers all parts of this course.
- ▶ It exams basic knowledge of machine learning.
- ▶ We will provide sample final exam for review.
- ▶ Time: 3:00pm - 5:00pm, December 19 (Saturday), 2026.
- ▶ Venue: TBD

Course Materials

- **Main material:** Lecture slides.
- **Recommended Reading books (not required):**
 - ▶ Abu-Mostafa, Y. S., Magdon-Ismail, M., & Lin, H. T. (2012). Learning from data. New York, NY, USA: AMLBook.
 - ▶ Bishop C., & Bishop H. (2024). Deep Learning: Foundations and Concepts. Springer.

Course Syllabus and Policies

You can find syllabus on Blackboard.

- ▶ Recommended reference books.
- ▶ Course summary.
- ▶ Homework information.
- ▶ Tutor and tutorial information.
- ▶ Project information.
- ▶ Grade distribution and final exam time.
- ▶ Course policies.

The first six parts are included in this lecture slides, please refer to the course syllabus for the **course policies**, e.g., policies on exam, homework submission, academic honesty, etc.

How to Study this Course

Lots of students worry about if they can pass this course. It should not be hard by doing the following things:

- ▶ **Math:** Try to understand basic math knowledge in the lecture.
- ▶ **Tutorials:** Try to attend the tutorials. Tutorial 1 introduces OpenCode and Codex; Tutorials 2-4 support the LLM course project.
- ▶ **Homework:** Discuss with classmates (but remember to write down everything using your own understanding), and **submit every homework**.
- ▶ **Project:** Start early, explore more possible ideas, and spend some time on writing the report.
- ▶ **Exam:** Understand most of the basic concepts included in the lecture, and **be sure to attend the exam**.